

Ask VJ

Data Warehouse Architecture Technical Reference Manual v2

Post-Migration Architecture | April 2026

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Chapter 1: Architecture Overview

1.1 Medallion Architecture (Post-Migration)

Ask VJ uses a three-layer medallion architecture:

- Bronze: 36 raw staging tables (Farvision + VJ Sales + VJOP)
- Silver: 19 tables (3 entity resolution + 16 dimensional model from developer's design)
- Gold: 8 views over Silver + 7 kept Farvision tables
- AI Layer: LLM queries Gold views to generate SQL from natural language

SOURCE SYSTEMS	BRONZE	SILVER	GOLD	AI LAYER
+-----+	+-----+	+-----+	+-----+	+-----+
Farvision	16 raw	Entity	Views	LLM SQL
(SQL Server)	tables	Resolution	over	from NL
+-----+			Silver	questions
VJ Sales	13 raw	Developer's	+	
(PostgreSQL)	tables	Dimensional	Farvision	
+-----+		Model:	tables	
VJOP Portal	7 raw	8 dims, 3 fact		
(SQL Server)	tables	+ LOV, xref		
+-----+	+-----+	+-----+	+-----+	+-----+

1.2 Source Systems

System	Database	Type	Key ID
Farvision ERP	VJDLIVE45	SQL Server	TenantId=75
VJ Sales App	PostgreSQL	Supabase	UUID IDs
VJOP Portal	RefferalAndLoyalty	SQL Server	INT IDs

1.3 Cross-System Join Keys

Key	Farvision	Other System	Confidence
BookingId	DimBookingMaster	VJ Sales: AllotmentPayment.bookingId	0.99
BookingId		VJOP: customer_bookings_units.fv_booking_	0.99
UnitId	DimUnitMaster	VJ Sales: Inventory.farvisionUnitId	0.95
BUIId	DimBusinessUnit	VJ Sales: Projects.buld	1.00
BUIId		VJOP: customer_bookings_units.BUIId	1.00
LedgerId	DimCustomerDetail	VJOP: customers.FV_LedgerID	0.95
LeadId	(none)	VJ Sales leadId <-> VJOP sales_app_lead_id	0.95

Chapter 2: Bronze Layer -- Raw Staging (36 Tables)

Append-only with audit metadata: _sync_id, _synced_at, _source_system, _batch_id.

2.1 Farvision ERP (16 Tables)

Table: stg_fv_dim_booking_master

Source: CRMG.DimBookingMaster

Column	Type	Description
BookingId	INT PK	Booking identifier
LedgerId	INT	Customer ledger
BUIId	INT	Universal project key
BookingDate	DATE	Booking date
NetBasicPrice	DECIMAL	Base price
IsCancelled	BOOLEAN	0=active
SalesPersonId	INT	Sales person
FiscalYearId	INT	56=FY25-26

Table: stg_fv_dim_business_unit

Source: ENGG.DimBusinessUnit

Column	Type	Description
BusinessUnitId	INT PK	Universal project key (BUIId)
BusinessUnit	VARCHAR	Project name
SegmentId	INT	Segment

Table: stg_fv_dim_receipt

Source: CRMG.DimReceipt

Column	Type	Description
ReceiptId	INT PK	Receipt ID (sic)
BookingId	INT	Linked booking
AmountLCY	DECIMAL	Amount
PaymentMode	VARCHAR	Cheque/NEFT/RTGS
DocumentDate	DATE	Payment date

Table: stg_fv_fact_duedate_outstanding

Source: CRMG.FactDueDatewiseOutstanding

Column	Type	Description
LedgerId	INT	Customer
Bill_Amount	DECIMAL	Total billed
Paid_Amount	DECIMAL	Total paid
Due_Amount	DECIMAL	Outstanding
DayAmt_15..More180	DECIMAL	7 aging buckets
Buld	INT	Project

2.2 VJ Sales App (13 Tables)

Timestamps are EPOCH MILLISECONDS. IDs are UUIDs. JSONB preserved.

Table: stg_vj_projects

Source: "Projects"

Column	Type	Description
projectId	UUID PK	Project ID
projectName	VARCHAR	Name
buld	INT	Maps to Farvision BUIId
reraNumber	VARCHAR	RERA

Table: stg_vj_inventory

Source: "Inventory"

Column	Type	Description
unitId	UUID PK	Unit ID
farvisionUnitId	INT	Cross-system key
totalCost	DECIMAL	Total price
BSP	DECIMAL	Base price/sqft

Table: stg_vj_leads

Source: "Leads"

Column	Type	Description
leadId	UUID PK	Lead ID
personId	UUID	Person ref
cpId	UUID	Channel partner
leadType	VARCHAR	walk-in/digital/CP

Table: stg_vj_allotment_payment

Source: "AllotmentPayment"

Column	Type	Description
allotmentPaymentId	UUID PK	Allotment ID
bookingId	INT	Cross-system key
status	VARCHAR	Payment Complete/Agreement Done
paidAmount	DECIMAL	Amount

2.3 VJOP (7 Tables)**Table: stg_rnl_customers**Source: *dbo.customers*

Column	Type	Description
id	INT PK	VJOP customer
FV_LedgerID	INT	Cross-system to Farvision

Table: stg_rnl_leadsSource: *dbo.leads*

Column	Type	Description
id	INT PK	VJOP lead
sales_app_lead_id	UUID	Cross-system to VJ Sales
referred_by	INT	Referrer customer ID
status	VARCHAR	Referral status

Chapter 3: Silver Layer -- Developer's Dimensional Model

19 tables total: 3 entity resolution (existing) + 16 from developer's relations.xlsx design.

All use UUID surrogate keys (_skey), DW audit columns, and centralized LOV lookups.

3.1 Entity Resolution (Existing)

Table: silver.project_crosswalk

Source: Auto-discovered from Bronze

Column	Type	Description
farvision_bu_id	INT UNIQUE	Universal project key
canonical_name	VARCHAR	Unified project name
vjsales_project_name	VARCHAR	VJ Sales name
vjop_project_name	VARCHAR	VJOP name

Table: silver.entity_map

Source: ID-based matching

Column	Type	Description
unified_customer_id	UUID	System-generated unified ID
farvision_booking_id	INT UNIQUE	Farvision BookingId
vjsales_lead_id	UUID UNIQUE	VJ Sales leadId
match_method	VARCHAR	booking_id_exact/unit_id_exact/lead_id_exact

3.2 Reference Tables

Table: silver.xref_map

Source: Centralized value standardization

Column	Type	Description
mapping_name	VARCHAR	e.g. pincode_to_city
from_domain	VARCHAR	Source domain
to_domain	VARCHAR	Target domain
from_value	VARCHAR	Raw value
to_value	VARCHAR	Standardized value

Table: silver.dim_lov

Source: Centralized status/type lookups (29 seed rows)

Column	Type	Description
lov_skey	UUID PK	Surrogate key
lov_type	VARCHAR	lead_status/lead_type/etc
lov_code	VARCHAR	Code
lov_value	VARCHAR	Display value

3.3 Dimension Tables

Table: silver.dim_calendar

Source: Generated 2015-2035

Column	Type	Description
calendar_skey	VARCHAR PK	YYYYMMDD format
calendar_dt	DATE	Actual date
fiscal_quarter	INT	Q1=Apr-Jun..Q4=Jan-Mar
fiscal_year	INT	Indian FY

Table: silver.dim_project

Source: stg_vj_projects + crosswalk

Column	Type	Description
project_skey	UUID PK	Surrogate key
bu_id	VARCHAR	Farvision BUID
project_name	VARCHAR	Project name
rera_number	VARCHAR	RERA
is_completed	BOOLEAN	Completion flag

Table: silver.dim_project_unit

Source: stg_vj_inventory

Column	Type	Description
project_unit_skey	UUID PK	Surrogate key
project_skey	UUID FK	-> dim_project
unit_id	UUID	VJ Sales unitId
fv_unit_id	VARCHAR	Farvision UnitId
wing_name	VARCHAR	Wing
unit_no	VARCHAR	Unit number
unit_type	VARCHAR	1BHK/2BHK/3BHK
total_cost_amt	NUMERIC	Total price
bsp_amt	NUMERIC	Base price/sqft

Table: silver.dim_buyer

Source: stg_vj_person + VJOP

Column	Type	Description
buyer_skey	UUID PK	Surrogate key
buyer_id	UUID	VJ Sales personId
name	VARCHAR	Full name
contact_number	VARCHAR	Mobile
email	VARCHAR	Email
pan	VARCHAR	PAN number
gender	VARCHAR	Gender

Table: silver.dim_employee

Source: stg_vj_users

Column	Type	Description
employee_skey	UUID PK	Surrogate key
employee_id	UUID	VJ Sales userId
employee_name	VARCHAR	Name
role_name	VARCHAR	Role

Table: silver.dim_channel_partner

Source: stg_vj_cp

Column	Type	Description
channel_partner_skey	UUID PK	Surrogate key
cp_id	UUID	VJ Sales cpId
billing_name	VARCHAR	Billing name
approval_status	VARCHAR	Status

3.4 Fact Tables**Table: silver.fact_lead**

Source: stg_vj_leads (20+ dimension FKs)

Column	Type	Description
fact_lead_skey	UUID PK	Surrogate key
lead_id	UUID UNIQUE	VJ Sales leadId
buyer_skey	UUID FK	-> dim_buyer
project_skey	UUID FK	-> dim_project
channel_partner_skey	UUID FK	-> dim_channel_partner
lead_dt_skey	VARCHAR FK	-> dim_calendar
lead_status_lov_skey	UUID FK	-> dim_lov (status)
lead_type_lov_skey	UUID FK	-> dim_lov (type)

Table: silver.fact_site_visit

Source: stg_vj_site_visits

Column	Type	Description
fact_site_visit_skey	UUID PK	Surrogate key
fact_lead_skey	UUID FK	-> fact_lead
project_skey	UUID FK	-> dim_project
employee_skey	UUID FK	-> dim_employee
mode	VARCHAR	Onsite/Video/F2F

Chapter 4: Gold Layer -- Business Views + Farvision Tables

Gold = thin business-facing layer. 8 views over Silver for LLM consumption.

7 kept Farvision tables (bookings, receipts, invoices, outstanding, inventory, referrals, funnel).

4.1 Views (pre-joined for LLM)

View	Description
gold.v_projects	Project name, bu_id, type, RERA, city, status
gold.v_units	Unit + project name, wing, floor, type, pricing, status
gold.v_buyers	Buyer name, phone, email, PAN, RM
gold.v_employees	Employee name, role, designation
gold.v_channel_partners	CP display ID, type, billing name, status
gold.v_leads	Pre-joined: buyer, project, CP, employees, LOV values
gold.v_site_visits	Pre-joined: lead, buyer, project, employee, CP, date
gold.v_conversion_rates	Lead-to-visit rate computed from Silver facts

4.2 Kept Farvision Tables

Table	Description
gold.fact_bookings	One row per booking. Upsert on farvision_booking_id.
gold.fact_receipts	Payments received. Upsert on farvision_receipt_id.
gold.fact_invoices	Demand letters. Upsert on farvision_invoice_id.
gold.snapshot_outstanding	Daily aging snapshot. 7 buckets (15-180+ days).
gold.snapshot_inventory	Daily inventory pricing snapshot.
gold.snapshot_referrals	Daily VJOP referral status snapshot.
gold.fact_daily_funnel_snapshot	Aggregated daily lead/booking counts.

Chapter 5: ETL Pipeline

Stage 1: EXTRACT (Bronze)

Farvision: 16 full-load queries (TenantId=75)
VJ Sales: 12 incremental (watermark on timestamp)
VJOP: 6 incremental (watermark on created_at)

Stage 2: RESOLVE (Silver)

Auto-discover projects via BUIId
Entity resolution: BookingId -> UnitId -> LeadId -> Fuzzy queue

Stage 2.5: PRE-GOLD QUALITY GATE

Check silver.project_crosswalk populated

Stage 3: TRANSFORM

Silver dims: calendar, country, project, project_unit, employee, CP, FOS, buyer
Silver facts: fact_lead (20+ FKs), fact_site_visit
Gold facts: bookings, receipts, invoices (Farvision)
Gold snapshots: outstanding, inventory, referrals

Stage 4: POST-GOLD VALIDATE

Orphans, nulls, negatives, entity coverage, completeness

Chapter 6: Data Lineage

6.1 Lead Lineage

```
VJ Sales "Leads" -> bronze.stg_vj_leads
-> silver.fact_lead (buyer/project/CP/employee/LOV lookups)
-> gold.v_leads (pre-joined view for LLM)
```

6.2 Site Visit Lineage

```
VJ Sales "SiteVisits" -> bronze.stg_vj_site_visits
-> silver.fact_site_visit (lead/project/employee lookups)
-> gold.v_site_visits (pre-joined view)
```

6.3 Booking Lineage (Farvision)

```
Farvision CRMG.DimBookingMaster -> bronze.stg_fv_dim_booking_master
-> silver.entity_map (BookingId match, conf 0.99)
-> gold.fact_bookings (agreement/registration/cancellation LATERAL joins)
```

6.4 Outstanding Lineage

```
Farvision FactDueDatewiseOutstanding -> bronze.stg_fv_fact_duedate_outstanding
-> gold.snapshot_outstanding (daily refresh, 7 aging buckets)
```

Chapter 7: Business Rules

7.1 Indian Fiscal Year

April-March. FY2026 = Apr 2025 - Mar 2026.

Q1=Apr-Jun, Q2=Jul-Sep, Q3=Oct-Dec, Q4=Jan-Mar.

7.2 Unit Status (Farvision)

Code	Status	Description
1	Sold	Booked/sold
2	Available	For sale
3	Blocked	Reserved

7.3 LOV Categories (dim_lov)

lead_status: New, Contacted, Follow Up, Site Visit Done, Negotiation, Booked, Lost, Cancelled

lead_type: CP, Walk-In, Digital, Referral, Self, Employee

lead_category: A, B, C (prioritization)

lead_response: Hot, Warm, Cold

7.4 Aging Buckets

Column	Range	Severity
day_amt_15	0-15 days	Recent
day_amt_30	16-30	1 month
day_amt_60	31-60	2 months
day_amt_90	61-90	3 months
day_amt_120	91-120	4 months
day_amt_180	121-180	6 months
day_amt_more_180	>180	Severe

7.5 DW Audit Columns (all Silver tables)

dw_load_ts: When record was first loaded (CURRENT_TIMESTAMP)

dw_update_ts: When record was last updated

dw_created_by: ETL process identifier ('etl_pipeline')

7.6 Surrogate Key Pattern

All Silver dimension/fact tables use UUID surrogate keys (gen_random_uuid()).

Named: {table}_skey (e.g. project_skey, buyer_skey, fact_lead_skey).

NAVL sentinel: md5('NAVL')::uuid for Not Available values.

NAPL sentinel: md5('NAPL')::uuid for Not Applicable values.